
Does Cost-Efficiency Travel? Rank Transfer and Frontier Stability in Fixed-Scaffold Agent Evaluation

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Abstract

1 Agent leaderboards increasingly report task success and dollar cost, inviting practitioners to
2 select a “cost-efficient” system from a single benchmark. We test whether this signal transfers
3 across workloads and whether any transfer reflects stable model-label cost propensity. Using
4 public Holistic Agent Leaderboard (HAL) results, we hold the displayed HAL Generalist
5 Agent scaffold fixed and compare repeated displayed model labels across six benchmarks.
6 Raw dollar-cost ranks transfer robustly in several high-overlap pairs: among pairs with at
7 least 12 shared labels, mean pairwise Spearman correlation is 0.84. A pooled log-cost model
8 with benchmark and displayed-label fixed effects explains 82 percent of observed log-cost
9 variance; after removing this stable observed label propensity, no high-overlap residual-cost
10 rank association is significant under a 20,000-draw permutation test. We then evaluate
11 aggregate CostPerSuccess, defined as dollar cost divided by $\max(\text{accuracy}, 1 \text{ percent})$: its
12 mean pairwise rank correlation falls to 0.59 across the same pairs. Thus raw cost transfer
13 largely encodes persistent label-level cost propensity, while success-adjusted efficiency is
14 less portable. Cost–accuracy frontier membership remains non-universal: no label tested on
15 at least five workloads lies on every frontier under nondominance, a HAL-inspired convex
16 hull, or a 5% near-tie sensitivity. Public data lack the token/provider fields required to isolate
17 historical list pricing or trace-level failure behavior.

18 1 Introduction

19 Agent benchmarks evaluate systems that reason over multiple steps, invoke tools, and interact with environments.
20 Reported behavior depends on the language model, scaffold, tools, prompts, execution limits, and task environ-
21 ment. Evaluation is also expensive: HAL reports 21,730 rollouts across nine benchmarks and roughly \$40,000 in
22 cost [Kapoor et al., 2026]. Cost-aware leaderboards therefore plot accuracy against dollar cost and recommend
23 cost–accuracy frontiers.

24 A practical question follows: if a displayed model label is efficient on one agent benchmark, is it a promising
25 low-cost choice elsewhere? This may hold when capability and resource use are portable, but can fail when
26 tool calls, retries, or stopping behavior differ by environment. We study this question descriptively, holding the
27 publicly displayed HAL Generalist Agent scaffold fixed and matching exact public model-label strings across
28 workloads.

29 Our design is coverage-aware. Most HAL scaffolds occur on one benchmark, making a global model–scaffold
30 decomposition unsupported. The Generalist Agent is the only broadly repeated scaffold. We retain six bench-
31 marks with adequate pairwise overlap and analyze accuracy rank, dollar-cost rank, aggregate CostPerSuccess
32 rank, and frontier membership only among shared labels. CostPerSuccess is defined as $C / \max(A, 0.01)$, where
33 A is accuracy as a fraction; the supplement reports a 5

34 2 Related work

35 Kapoor et al. [2026] introduce HAL as a cost-aware leaderboard across coding, web, science, and customer-
36 service tasks. Kapoor et al. [2024] argue for joint accuracy–cost evaluation of agents. SWE-Effi studies resource-
37 constrained software-engineering agents and reports scaffold–model interaction within one domain [Fan et al.,

Table 1: Higher-overlap pairs. Brackets are configuration-resampling sensitivity intervals; ND is nondominated-frontier Jaccard similarity on the shared-label cohort.

Pair	N	Accuracy ρ	Cost ρ	ND Jaccard
CORE-Bench Hard–GAIA	14	0.60 [0.16, 0.81]	0.93 [0.72, 0.99]	0.43
CORE-Bench Hard–SWE-bench Mini	14	0.48 [-0.05, 0.82]	0.74 [0.29, 0.93]	0.25
CORE-Bench Hard–TAU Airline	12	0.50 [-0.12, 0.86]	0.78 [0.26, 0.98]	0.17
GAIA–SWE-bench Mini	17	0.71 [0.31, 0.91]	0.89 [0.61, 0.99]	0.33
GAIA–TAU Airline	14	0.58 [0.14, 0.85]	0.86 [0.58, 0.97]	0.29
SWE-bench Mini–TAU Airline	14	0.76 [0.40, 0.91]	0.82 [0.46, 0.97]	0.43

2025]. Ndzomga [2026] study within-benchmark rank preservation under task reduction, while Benchmark² studies cross-benchmark rank consistency for static LLMs [Qian et al., 2026]. None directly establishes cross-workload rank transfer for interactive-agent dollar costs under a repeated scaffold.

3 Data and method

We analyze the public `all_leaderboards_costs_HAL.csv` snapshot distributed with Ndzomga [2026]: 242 displayed evaluations across nine benchmarks, 11 scaffolds, and 29 model labels. Accuracy and total dollar cost are complete, but 235 rows report one run. The frozen source, exact hash, and code are in the anonymous supplement.

The primary cohort fixes the displayed HAL Generalist Agent scaffold: 21 labels on CORE-Bench Hard, 17 on GAIA, 9 on SciCode, 7 on ScienceAgentBench, 18 on SWE-bench Verified Mini, and 14 on TAU-bench Airline. USACO has one Generalist result and is excluded. For each pair with at least five shared labels, we compute Spearman ρ and Kendall rank correlation for accuracy and dollar cost. We use 10,000 configuration-resampling draws and 10,000 label-pairing permutations; these quantify finite-label sensitivity, not rollout-level or population-model uncertainty. Holm–Bonferroni correction is applied separately to 15 accuracy and 15 cost tests.

A nondominated point has no competitor that is at least as accurate and no more costly with one strict improvement. We also implement a HAL-inspired origin-anchored convex-hull reconstruction and a 5% tolerance sensitivity where dominance requires 5% lower cost and 5% higher accuracy. The tolerance is a near-tie stress test, not a calibrated measurement-error model.

4 Results

4.1 Raw cost-rank transfer is explained by stable label cost propensity

The robust raw correlations invite a confound check: cost rankings may encode persistent displayed-label price/cost tiers rather than portable execution dynamics. The frozen CSV lacks token components, provider alias, and historical price-card fields, so we cannot claim a list-price adjustment. We fit an observational pooled log-cost model on 86 Generalist rows with benchmark and displayed-label effects. It explains 82.0 percent of log-cost variance, with adjusted R-squared of 72.7 percent. The six high-overlap raw correlations collapse after residualization: none of the residual-cost rank correlations is significant under a 20,000-draw permutation test. Thus raw cost transfer largely reflects stable observed label-level cost propensity. This propensity may combine provider price tier, reasoning configuration, token use, cache treatment, routing, and unrecorded setup; it is not identifiable as list price from the public CSV. Exact adjusted coefficients are reported in the anonymous supplement. The raw cross-workload rank-transfer matrix, including each pair’s Spearman correlation and shared-label count, is provided in the anonymous supplement. The raw dollar-cost structure is largely absorbed by stable displayed-label cost propensity.

4.2 Success adjustment reduces portability

Raw dollar cost does not answer which label buys expected success most cheaply. With the pre-specified one-percent denominator floor, mean CostPerSuccess rank correlation across the six high-overlap pairs is 0.59, versus 0.84 for raw cost; five of six pairs decline. The largest changes occur for GAIA–SWE-mini (0.89 to 0.29), CORE–SWE-mini (0.74 to 0.32), and SWE-mini–TAU (0.82 to 0.45); CORE–TAU is the exception. Alternative zero-exclusion and 5

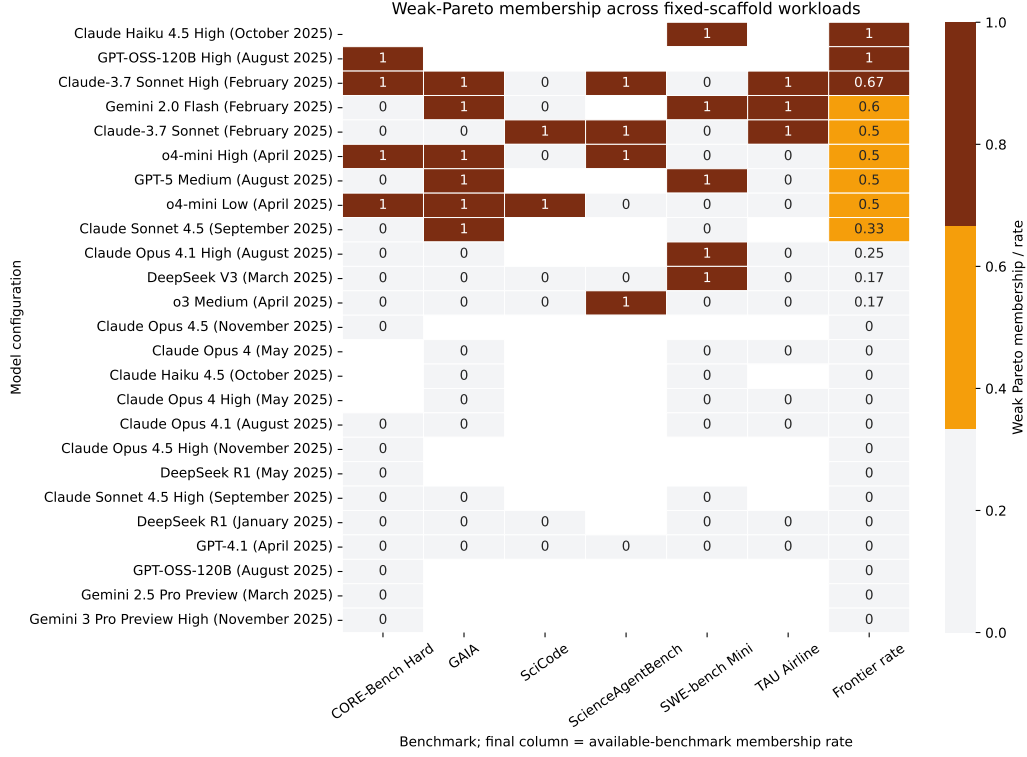


Figure 1: Nondominated cost–accuracy membership across primary workloads. The final column is each label’s rate over available workloads. No broadly tested label appears on every available frontier.

4.3 Held-out portability is benchmark-specific

In leave-one-benchmark-out prediction, displayed-label cost propensity learned elsewhere predicts raw cost ranks for CORE ($\rho = 0.78$, $N = 14$, $p = 0.0016$), GAIA (0.87 , $N = 16$, $p < 0.0001$), SWE-mini (0.90 , $N = 16$, $p < 0.0001$), and TAU (0.83 , $N = 14$, $p = 0.0004$); it does not predict SciCode (-0.17 , $N = 9$) or ScienceAgentBench (-0.64 , $N = 7$). This is benchmark-specific heterogeneity, not a domain claim. The full plot is in the supplement.

4.4 Frontier membership is non-universal

No label tested on at least five workloads lies on every cost–accuracy frontier. This conclusion survives nondominance, the HAL-inspired hull, and 5% near-tie tolerance. The stronger statement that no label appears on more than 4/6 frontiers does not survive: under tolerance, o4-mini Low appears on 5/6. We therefore retain only the robust non-universal-winner conclusion.

4.5 Scaffold and frontier definitions matter

The source HAL labels recover 30 rows under the hull reconstruction, with two discrepancies; ordinary nondominance includes 19 additional points. A discrete buyer choosing one system and a deployer willing to randomize between systems therefore face different decision frontiers. Within individual benchmarks, scaffolds also differ substantially: across 15 shared labels on SWE-mini, displayed SWE-Agent averages 27.4 percentage points higher accuracy than the Generalist scaffold, with higher median cost. These contrasts motivate fixing the scaffold; they are not a general causal scaffold estimate.

5 Discussion and threats to validity

A single benchmark can supply a useful prior about *raw* observed dollar cost for some held-out workloads, but the analysis does not establish portable workload-specific execution-cost dynamics: raw cost-rank transfer is largely absorbed by stable displayed-label cost propensity, and CostPerSuccess is less portable. Because this propensity may include provider price tier, reasoning configuration, token volume, cache treatment, routing,

100 and unrecorded setup, it cannot be reduced to a list-price claim from the frozen public CSV. Practitioners
101 should validate success-adjusted cost and token behavior on representative target tasks rather than treating one
102 leaderboard frontier as a universal efficiency mechanism.

103 Construct validity is limited because public `Model`s and `Agent Name` strings do not fully encode prompts,
104 tools, budgets, harness versions, or pricing assumptions. The CSV also lacks provider aliases and token
105 components, precluding historical list-price or common-tariff reconstruction; public trace archives are encrypted.
106 `CostPerSuccess` is an aggregate dollars-per-expected-success proxy, not task-level successful-run cost; the 1

107 6 Conclusion

108 Raw dollar-cost ranks transfer robustly across several high-overlap public HAL workload pairs, but pooled
109 label and benchmark cost propensity explains most of this signal. Aggregate `CostPerSuccess` is less portable,
110 showing that predictable cheapness is not equivalent to predictable dollars per expected success. The public CSV
111 does not establish portable workload-specific execution-cost dynamics or isolate historical provider list prices.
112 Cost-accuracy frontier membership is nevertheless non-universal under multiple decision-frontier definitions.
113 Raw dollar-cost leaderboards should therefore be interpreted as a mixture of persistent label-level cost propensity
114 and workload behavior, then validated with success-adjusted and trace-aware measures on target tasks.

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131 Answer: [Yes]

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136 Question: Does the paper discuss the limitations of the work performed by the authors?

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